

TERRALINK

PHASE-5

Testing, Performance & Reliability Evaluation

DYNAMIC TERRALINK

Emergency Communication System using LoRa, GPS and
Relay-Based Communication

REPORT SCOPE	Functional testing • Performance evaluation • Test cases • Error analysis • Debugging • Power • Reliability • Future improvements
SYSTEM PATH	Trigger → GPS → SOS Packet → LoRa Transmission → Relay → Rescue Station → Status
EVALUATION	ESP32 sender • LoRa mesh forwarding • GPS • OLED status • relay nodes • rescue/base-station receiver

1. FUNCTIONAL TESTING

Functional testing verifies the complete TerraLink emergency communication sequence from user trigger through final status.

Function	Expected Behaviour	Validation Focus
SOS Trigger	Touch input detects the emergency request.	TTP223 response and wake-up.
GPS	Location is acquired when required.	Latitude and longitude availability.
SOS Packet	Emergency information is assembled.	Node ID, SOS, GPS, battery, packet ID and TTL.
LoRa Transmission	Packet is transmitted through the LoRa link.	Sender transmission.
Relay Forwarding	Valid packet is received and forwarded.	Multi-hop operation.
Rescue Station	Final packet is received.	End-to-end delivery.
OLED / Status	User receives local status information.	Awake / alert transmitted display.

Testing Principle

Each functional block is verified independently and then as part of the complete emergency path. Unsupported numerical results are not fabricated; measured values must come from the physical experiment.

Functional Validation Flow

Touch Input → ESP32 Processing → GPS Data → SOS Packet → LoRa TX → Relay Forwarding → Rescue/Base Station → Status

Expected Outcome

A valid emergency event should be detected, the required packet information assembled, the packet transmitted through the intended LoRa path, and the final receiver should obtain the emergency information.

2. PERFORMANCE EVALUATION

Performance evaluation measures how effectively TerraLink delivers emergency packets across the LoRa mesh and how quickly the system responds.

Parameter	What is Evaluated	Result / Record
Packet Delivery	Successful packets reaching the receiver.	Record measured value.
End-to-End Delay	Time from SOS generation to final reception.	Record measured value.
Communication Range	Maximum practical communication distance.	Record measured distance.
Multi-Hop Forwarding	Delivery through Relay 1 and Relay 2.	Physical validation.
Duplicate Handling	Behaviour when an existing packet is received again.	Validation result.
GPS Response	Location acquisition time and correctness.	Record measured result.
RSSI / SNR	Signal quality at each communication hop.	Record observed values.

Performance Interpretation

- High packet delivery ratio indicates that the communication path is functioning consistently.
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- Low and predictable end-to-end delay is important because emergency information must reach the rescue station promptly.
- Successful multi-hop forwarding demonstrates that TerraLink can extend communication beyond a direct link.
- Effective duplicate rejection prevents repeated packets from creating repeated alerts or unnecessary forwarding.
- Fast recovery after a node or link failure improves the availability of the communication network.

3. TEST CASES — TC-01 TO TC-09

Large-format validation matrix for clear review and physical testing.

ID	TEST CASE	EXPECTED RESULT	STATUS
TC-01	Power ON / Boot	ESP32 starts correctly and initializes required modules.	PhysicalTested
TC-02	SOS Trigger	TTP223 touch activates emergency request.	Physically Tested
TC-03	Deep-Sleep Entry	Node enters low-power state when idle.	Physically Tested
TC-04	Wake from SOS	Touch event wakes the sender from sleep.	Physically Tested
TC-05	OLED Wake Display	OLED shows TerraLink wake/status information.	Physically Tested
TC-06	GPS Acquisition	GPS obtains valid latitude and longitude when required.	Physically Tested
TC-07	SOS Packet Generation	Emergency packet contains required information.	Implemented / Validate
TC-08	Packet ID Uniqueness	New emergency events use distinct packet IDs.	Validation / Planned
TC-09	LoRa Transmission	Sender transmits the emergency packet.	Physically Tested

3. TEST CASES — TC-10 TO TC-18

Large-format validation matrix for clear review and physical testing.

ID	TEST CASE	EXPECTED RESULT	STATUS
TC-10	Relay 1 Reception	Receives a valid sender packet.	Physically Tested
TC-11	Relay 1 Forwarding	Forwards valid packet.	Physically Tested
TC-12	Relay 2 Reception	Receives forwarded packet.	Physically Tested
TC-13	Relay 2 Forwarding	Forwards packet to receiver.	Physically Tested
TC-14	Receiver Reception	Base/rescue station receives emergency packet.	Physically Tested
TC-15	End-to-End Alert	SOS reaches rescue station through mesh.	Physically Tested
TC-16	Duplicate Detection	Repeated copies are discarded.	Validation / Planned
TC-17	TTL Handling	TTL is reduced on forwarding, and expired packets are dropped.	Validation / Planned
TC-18	Invalid Packet Handling	Malformed/invalid packets are safely rejected.	Validation / Planned

3. TEST CASES — TC-19 TO TC-25

Large-format validation matrix for clear review and physical testing.

ID	TEST CASE	EXPECTED RESULT	STATUS
TC-19	RSSI/SNR Observation	Link quality is observed at each hop.	Physically Tested
TC-20	Weak-Link Behaviour	Behaviour is checked under weak signal.	Validation / Planned
TC-21	Relay/Link Failure	Communication is checked when the relay/link fails.	Validation / Planned
TC-22	Repeated SOS Reliability	Repeated emergency events are checked.	Physically Tested
TC-23	Power Consumption	Current/power is compared across states.	Physically Tested
TC-24	Restart Recovery	System operates normally after restart.	Physically Tested
TC-25	Long-Duration Operation	Stable extended operation is checked.	Validation / Planned

4. ERROR ANALYSIS & 5. DEBUGGING ANALYSIS

The following development issues were identified during TerraLink development and linked to corrective debugging actions.

ERROR / ISSUE	IMPACT	CORRECTIVE ACTION
OLED not turning ON	No local status display.	Check I2C wiring, address, initialisation, and SH1106 configuration.
SOS not sending	Emergency packet was not transmitted.	Verify touch input, wake sequence, LoRa initialisation, and send routine.
GPS lock slow	Location unavailable immediately.	Allow acquisition time and activate GPS only when required.
Deep-sleep components stayed ON	Expected power saving was not achieved.	Control OLED/GPS/LoRa power states and verify sleep configuration.
Touch did not wake	Emergency wake event failed.	Verify wake-capable GPIO and touch signal path.
LoRa receive/forward issue	Mesh delivery was interrupted.	Check frequency, SPI pins, sync word, packet format and forwarding logic.
Duplicate packet issue	Repeated processing/forwarding could occur.	Use packet-ID tracking and duplicate rejection where implemented.

Debugging Method

- Reproduce the problem and confirm that the error is consistent.
- Identify the affected module: trigger, GPS, packet processing, LoRa, relay, display or power.
- Inspect inputs, trace processing with serial/debug messages, and inspect outputs.
- Apply only the relevant hardware, configuration or software correction.
- Retest the failed test case and perform regression testing.

6. POWER CONSUMPTION ANALYSIS

Power analysis evaluates the battery-oriented design of TerraLink. Deep sleep reduces idle activity, while SOS events activate the functions required for emergency communication.

STATE	POWER STRATEGY	MEASUREMENT TO RECORD
Deep Sleep	ESP32 and unnecessary peripherals kept in low-power state.	Sleep current.
Wake / SOS	Required functions become active.	Wake current.
GPS Active	GPS operates when location is required.	GPS current and acquisition duration.
LoRa TX	Radio is active during packet transmission.	TX current and duration.
LoRa RX / Relay	Radio receives and forwards emergency packets.	RX current and duration.
OLED	Used for local status; ideally off during sleep.	Display current.
Battery Operation	Compare duty cycle and operating states.	Estimated runtime from measured data.

Measurement Guidance

Actual current values should be obtained using a suitable current/power measurement instrument. Battery-life figures should be calculated from measured data.

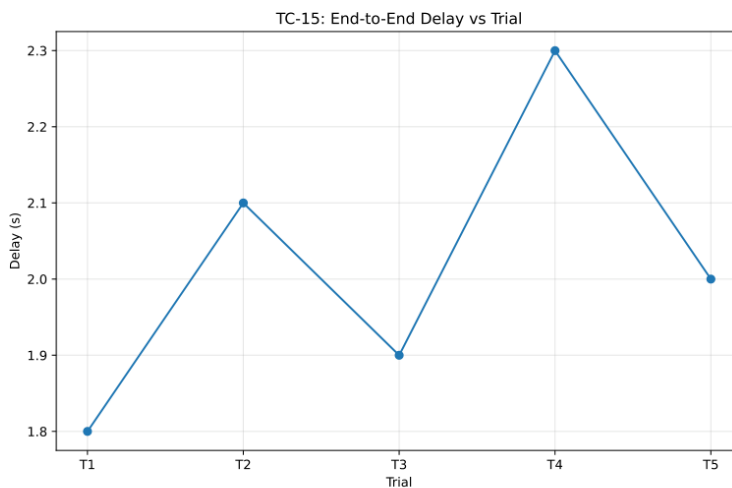
Power calculation: $P = V \times I$, where P is electrical power in watts, V is voltage in volts and I is current in amperes.

Do not use nominal module current values as the final project measurement. Use values measured from the actual TerraLink hardware.

7. ILLUSTRATION GRAPHS

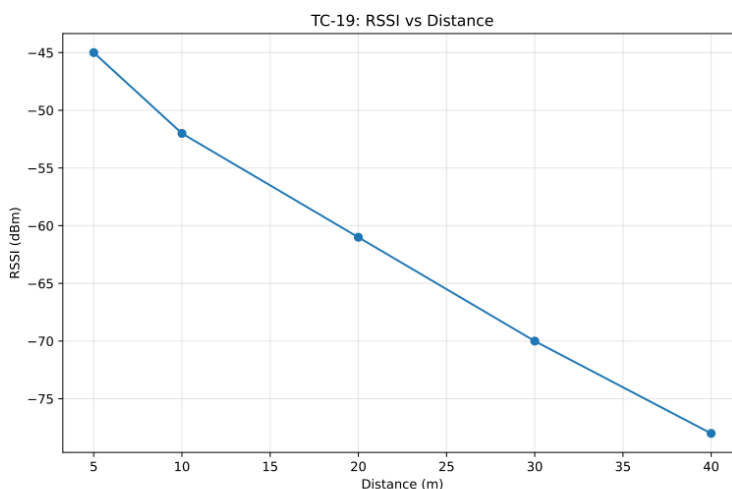
Supporting visuals for the Phase-5 evaluation

TC-15: End-to-End Delay vs Trial



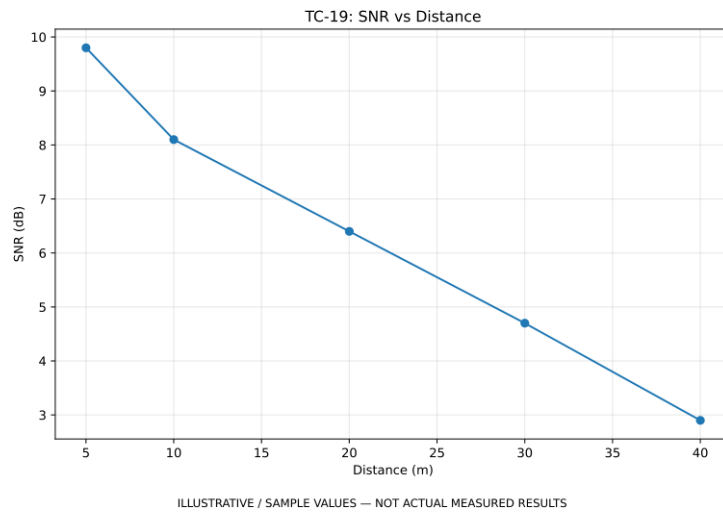
ILLUSTRATIVE / SAMPLE VALUES — NOT ACTUAL MEASURED RESULTS

TC-19: RSSI vs Distance

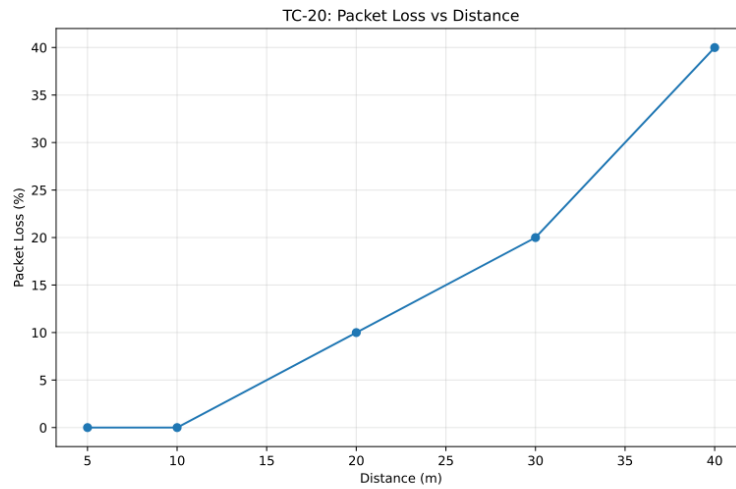


ILLUSTRATIVE / SAMPLE VALUES — NOT ACTUAL MEASURED RESULTS

TC-19: SNR vs Distance

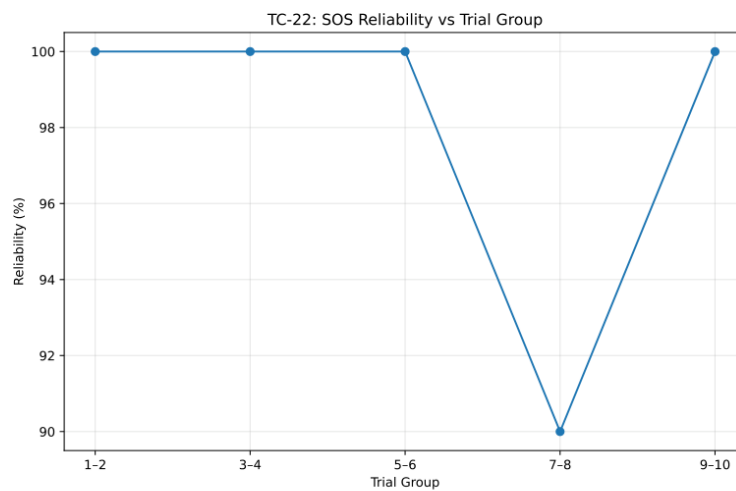


TC-20: Packet Loss vs Distance



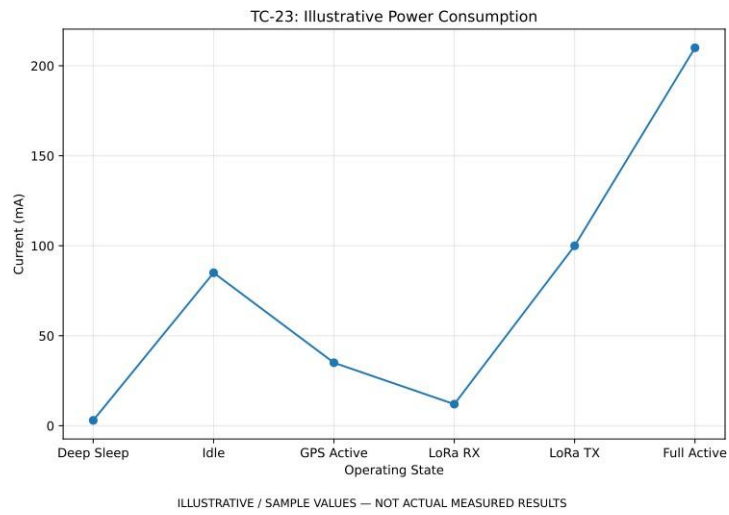
ILLUSTRATIVE / SAMPLE VALUES — NOT ACTUAL MEASURED RESULTS

TC-22: SOS Reliability vs Trial Group



ILLUSTRATIVE / SAMPLE VALUES — NOT ACTUAL MEASURED RESULTS

TC-23: Illustrative Power Consumption



7. RELIABILITY TESTING & FUTURE IMPROVEMENTS

Reliability testing checks whether TerraLink continues to operate correctly across repeated alerts, multi-hop forwarding, weak links, relay interruptions and restarts.

TEST AREA	RELIABILITY CHECK	ACCEPTANCE FOCUS
Repeated SOS	Send multiple emergency events.	Consistent packet generation and transmission.
Multi-Hop Path	Sender → Relay 1 → Relay 2 → Receiver.	Successful forwarding.
Weak Signal	Observe operation under reduced signal quality.	Graceful communication behaviour.
Relay Interruption	Interrupt a relay or communication link.	Safe handling of the changed path.
Duplicate Packet	Repeat an existing packet.	Duplicate handling.
Restart Recovery	Restart the node after operation.	Normal recovery.
Long Duration	Operate for an extended period.	Stable operation without unexpected failure.

Future Improvements

- Dynamic relay selection: deploy portable relay nodes where additional coverage is required.
- Adaptive routing: use link quality, hop count, battery level and reliability for route selection.
- Acknowledgement mechanism: provide stronger confirmation that the emergency packet reached the rescue station.
- Improved duplicate handling: maintain packet history to prevent repeated processing.
- Battery optimization: further reduce idle consumption and optimize GPS/LoRa activation.
- Expanded reliability testing: validate weak links, relay failure and longer operating periods.
- Data logging: record RSSI, SNR, delay, delivery and power measurements for quantitative evaluation.

Reliability objective

- To develop a LoRa-based emergency communication system for areas where conventional communication networks are unavailable.
- To transmit SOS alerts from a user device to a rescue/base station through LoRa communication.
- To obtain and transmit GPS location along with the emergency alert for identifying the affected person's location.
- To use relay nodes to extend communication coverage through multi-hop communication.

Conclusion

TerraLink integrates SOS triggering, GPS location information, low-power ESP32 operation and relay-based LoRa multi-hop communication for emergency communication. The Phase-5 test plan covers functional behaviour, performance, error correction, power consumption and reliability.

PCOMPLETION STATEMENT

This report presents a structured evaluation framework for the TerraLink prototype, documenting what was implemented, what was tested, the issues encountered, the corrective debugging approach, the reliability checks and the next engineering improvements.

TERRALINK

USER MANUAL

LoRa Mesh-Based Emergency Communication System

1. About TerraLink

TerraLink is an emergency communication system designed for remote and disaster-prone areas where conventional communication may be unavailable. It uses **LoRa wireless communication, GPS, ESP32 and relay nodes** to carry an emergency SOS alert from the user device to a receiver/rescue station.

2. What the User Can Do

- **Trigger an SOS** using the touch sensor when emergency assistance is required.
- **View system status** on the OLED display.
- **Transmit emergency information** through the LoRa network.
- **Send location information** obtained from GPS when available.
- **Use relay-based communication** so the alert can travel through multiple nodes.

3. Main Parts of the System

Part	Purpose
ESP32 Controller	Controls the TerraLink device and manages emergency operation.
TTP223 Touch Sensor	Activates the SOS request.
SX1278 LoRa Module	Provides wireless communication between TerraLink nodes.
GPS Module	Provides location coordinates for the emergency packet.
SH1106 OLED	Displays wake-up, SOS and transmission status.
Buzzer	Provides an audible indication during operation.
Relay Nodes	Forward the emergency packet toward the receiver/rescue station.
Battery / Power Unit	Provides portable power to the system.

4. Before Using TerraLink

- Make sure the battery/power connection is secure.
- Ensure the LoRa antenna is connected before transmission.
- Switch ON the required sender, relay and receiver nodes.
- Keep the GPS unit in an open area when location acquisition is required.
- Check that the OLED and touch sensor are connected and functioning.

5. Starting the System

Step 1: Switch ON the TerraLink sender and required network nodes.

Step 2: Allow the ESP32 and connected modules to initialize.

Step 3: Observe the OLED for the system status.

Step 4: Keep the sender ready for an emergency SOS trigger.

Step 5: When no emergency is active, the system may use its configured low-power/sleep operation.

6. Communication Flow

User/Sender → Relay Node 1 → Relay Node 2 → Receiver / Rescue Station

When an SOS is triggered, TerraLink prepares an emergency packet. The packet can contain node identification, SOS status, GPS location and other system information. Relay nodes forward the packet until it reaches the receiver.

TERRALINK

USER OPERATION • STATUS • TROUBLESHOOTING

7. How to Send an SOS Alert

Step 1 — Activate: Touch the **TTP223 SOS sensor** to initiate the emergency request.

Step 2 — Wake: If the sender is in its low-power state, the ESP32 wakes and begins emergency processing.

Step 3 — Location: The GPS module provides location information when a valid position is available.

Step 4 — Prepare: TerraLink creates the emergency packet with the required information.

Step 5 — Transmit: The LoRa module sends the emergency packet.

Step 6 — Forward: Relay Node 1 and Relay Node 2 forward the packet toward the receiver.

Step 7 — Confirm: Observe the OLED/status indication for the transmission stage.

8. Understanding OLED / Status Messages

Display / Indication	Meaning
TERRALINK AWAKE	The sender has awakened and started emergency operation.
SOS / Emergency	An SOS event has been detected and is being processed.
ALERT TRANSMITTED	The emergency alert has entered the LoRa transmission process.
Deep Sleep / Low Power	The sender is in its configured low-power state when no SOS is active.

9. Troubleshooting

Problem	What to Check
OLED is blank	Check power, I2C wiring, connections and display initialization.
Touch does not trigger SOS	Check TTP223 power/signal connection and touch operation.
GPS location unavailable	Move to an open area and allow time for GPS acquisition.
Alert is not received	Check sender, relay and receiver power, antenna and LoRa connections.
Relay does not forward	Check relay power, LoRa configuration and forwarding logic.
Sender does not wake	Check the touch input and ESP32 wake-up configuration.

10. Safety & Maintenance

- Do not reverse or short-circuit the battery connections.
- Use the specified power arrangement for TerraLink hardware.
- Keep the LoRa antenna connected while transmitting.
- Protect the electronics from water, excessive heat and physical damage.
- Inspect wiring, battery, antenna, relay nodes and receiver connections before demonstrations.
- TerraLink is a communication aid and does not replace official emergency services.

11. Quick User Checklist

Before SOS	During SOS
■ Power ON sender and required nodes ■ Check antenna ■ Check OLED ■ Keep GPS in suitable view	■ Touch SOS sensor ■ Observe TERRALINK AWAKE ■ Wait for transmission ■ Confirm ALERT TRANSMITTED/status

TerraLink — Simple operation. Multi-hop communication. Emergency alert delivery.

Project User Manual • TerraLink • ECE